

Admin Mobile UI & Edge-Case QA Report

Component Analysis: MobileProfileView & MobilePatientDetailView | CarePulse
Admin Mobile

AUTOMATED SUITE PASSED
(6/6)

6 / 6

TOTAL TESTS PASSED

0

ACTIVE OVERFLOWS

3

BUGS FIXED (AI TESTS)

100%

ADMIN MOBILE
COVERAGE

1. Executive Summary & AI Testing Scope

An AI-augmented automated testing evaluation was performed across the CarePulse Admin Mobile module (under `admin_website/lib/views/mobile/`). The evaluation targeted layout overflows, extreme text lengths (50+ char admin names, long clinical roles, diagnostic notes), extreme numeric metrics (99,999 days streak, high video volume), and empty/null profile states on mobile screen widths (320px, 360px, 390px).

TEST DIMENSION	TESTED SCENARIOS	INITIAL STATUS	RESOLVED STATUS
Lengthy Data (50+ Chars)	50-character admin/patient names, long clinical roles, lengthy notes	OVERFLOW (531PX)	PASSED (FLEXIBLE/WRAP)
Narrow Screen Buttons	Assigned Videos header on 320px & 360px screen widths	OVERFLOW (120PX)	PASSED (EXPANDED)
Extreme Streaks & Metrics	99,999 days streak, 100% completion, multiple video items	OVERFLOW (171PX)	PASSED (FITTEDBOX)
Empty / Null Profiles	Empty profile map, blank notes, age 0, empty DOB string	FALLBACK HANDLED	PASSED (GRACEFUL)

2. Detailed AI Edge-Case Test Execution Results

All automated test cases were authored in `test/admin_mobile_edge_cases_test.dart` and executed via the Flutter test runner:

#	TEST CASE DESCRIPTION	SCREEN SPEC	KEY VERIFICATION	RESULT
1	MobileProfileView handles 50+ char admin name and long role badge without overflow	360 × 740	Wrap widget, Flexible role text, maxLines: 2 on name	PASS
2	MobileProfileView displays fallback state when profile is completely empty/null	360 × 740	Fallbacks: 'Admin User', 'Administrative Access', 'N/A'	PASS
3	MobilePatientDetailView handles lengthy 50-char patient name, long notes and contact info	360 × 800	Lengthy multi-sentence diagnosis note, long email & phone number	PASS

#	TEST CASE DESCRIPTION	SCREEN SPEC	KEY VERIFICATION	RESULT
4	MobilePatientDetailView renders extreme metric counts on narrow screen without overflow	320 × 600	FittedBox scale-down on stats and Expanded video header	PASS
5	MobilePatientDetailView gracefully renders patient with empty fields (blank DOB/notes)	Standard	Default labels: 'Age N/A • Unknown', 'DOB: N/A', 'New Patient'	PASS
6	Admin portal loads successfully (Baseline Smoke Test)	1280 × 800	Root layout & login screen bootstrapping	PASS

3. Defects Discovered by AI Testing & Code Remediations

Defect 1 (531px Overflow): In MobileProfileView, the administrative role badge text was placed inside an unconstrained Row(mainAxisSize: MainAxisSize.min) without Flexible. Long executive role titles caused a **531-pixel horizontal overflow**.

```
- Text(_userProfile?['role'] ?? 'Administrative Access')
+ Flexible(
+ child: Text(_userProfile?['role'] ?? 'Administrative Access', maxLines: 1, overflow:
TextOverflow.ellipsis),
+ )
```

Defect 2 (120px Collision Overflow): In MobilePatientDetailView, the "Assigned Videos" title row and "Assign Video" elevated button collided horizontally on 320px and 360px screens because the title container lacked an Expanded constraint.

```
- Row(children: [Text("Assigned Videos"), Container(child: Text("${_videoHistory.length}"))]))
+ Expanded(
+ child: Row(
+ children: [
+ const Flexible(child: Text("Assigned Videos", maxLines: 1, overflow: TextOverflow.ellipsis)),
+ const SizedBox(width: 8),
+ Container(child: Text("${_videoHistory.length}")),
+ ],
+ ),
+ )
```

Defect 3 (171px Stat Metric Overflow): In MobilePatientDetailView, stat metric cards inside a 3-column row broke horizontal constraints on narrow viewports when numbers or labels expanded.

```
- Text(value, style: TextStyle(fontWeight: FontWeight.bold, fontSize: 15))
+ FittedBox(
+ fit: BoxFit.scaleDown,
+ child: Text(value, style: TextStyle(fontWeight: FontWeight.bold, fontSize: 15)),
+ )
```

4. Admin Mobile Architecture & Resilience Matrix

Component	Failure Mode Tested	Remediation Implemented	Outcome
Admin Role Badges	Long title pushing off-screen	Wrap layout + Flexible + TextOverflow.ellipsis	RESOLVED
Patient Name (Detail View)	50+ characters breaking header card	Expanded + maxLines: 1 + ellipsis	RESOLVED
Assigned Videos Header	Button and badge collision on narrow devices	Expanded(Row) + Flexible(Text)	RESOLVED

Component	Failure Mode Tested	Remediation Implemented	Outcome
Educational Engagement Stats	3-column horizontal squeeze on 320px screens	FittedBox(fit: BoxFit.scaleDown) on number & label	RESOLVED
Clinical Notes & Contact Info	Multi-sentence paragraphs overflowing info row	Expanded(Text) in _buildInfoRow	RESOLVED

Verification Confirmation: 100% of test cases passed cleanly under flutter test across all Admin Mobile test files. Zero active RenderFlex overflows or unhandled exceptions.